

Model-based estimation of marine mammal bycatch

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Model statement

For a given marine mammal stock and fishery:

$$y_i \sim \text{Binom}(N_i, p_i)$$

$$p_i = \frac{1}{W_i} \sum_j w_{ij}$$

$$N_i \sim \text{Pois}(\mathbf{W}_i \lambda_i)$$

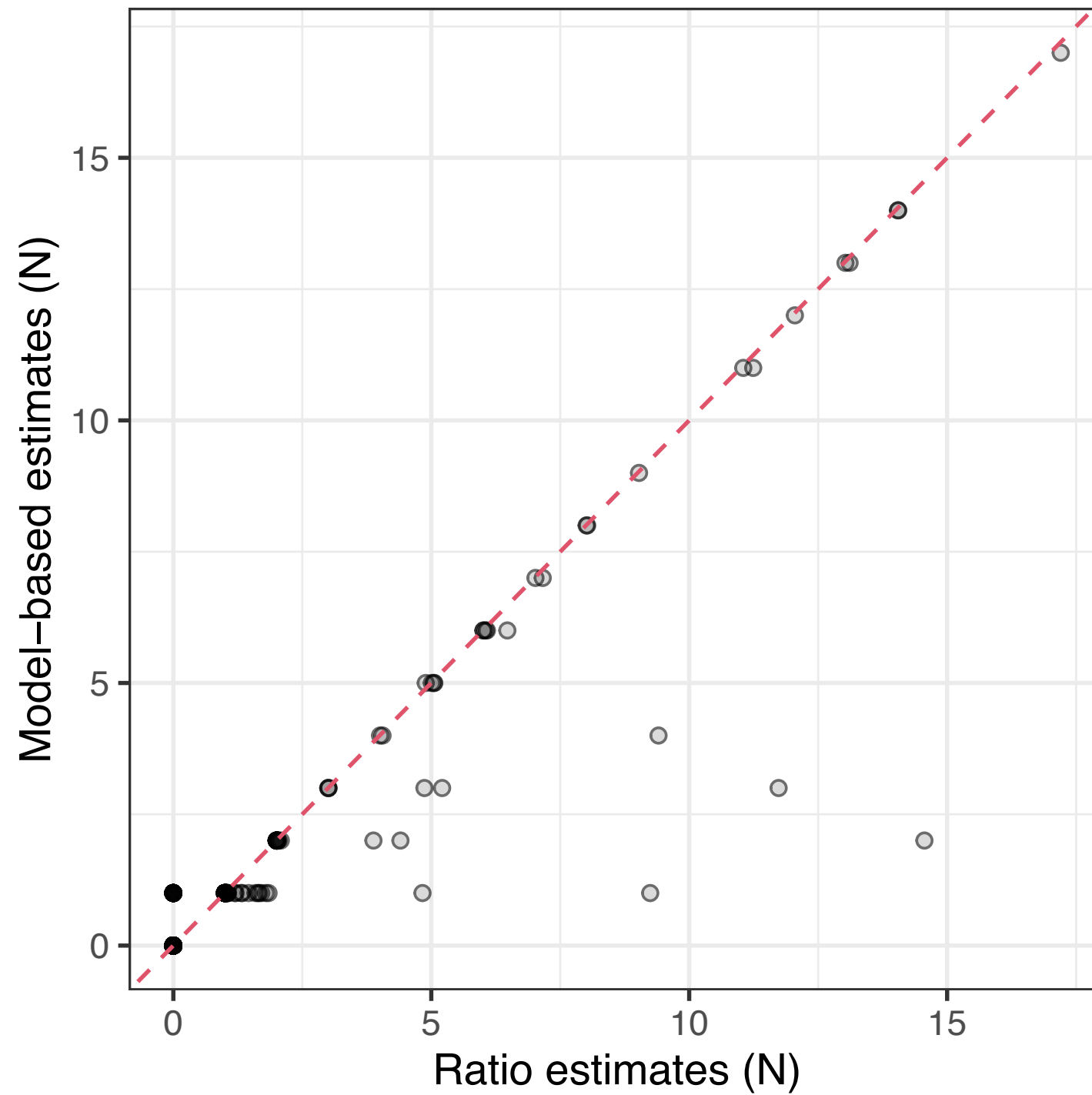
$$\log(\lambda_i) = \mathbf{x}_i' \boldsymbol{\beta} + \epsilon_i$$

$$\log(w_{ij}) \sim \mathcal{N}(\mu_i, \sigma_i^2)$$

$$\text{Strata } (i) \left\{ \begin{array}{l} \text{Week: } [1, 18), [18, 36), [36, 52) \\ \text{Vessel length: } [0, 60), [60, 125), 125+ \\ \text{Area: Bering Sea, Aleutian Island, Gulf of Alaska} \\ \text{Year: 2012–2023} \end{array} \right.$$

$$\text{Haul } (j): 1, 2, \dots, J$$

Model-based vs. ratio estimates



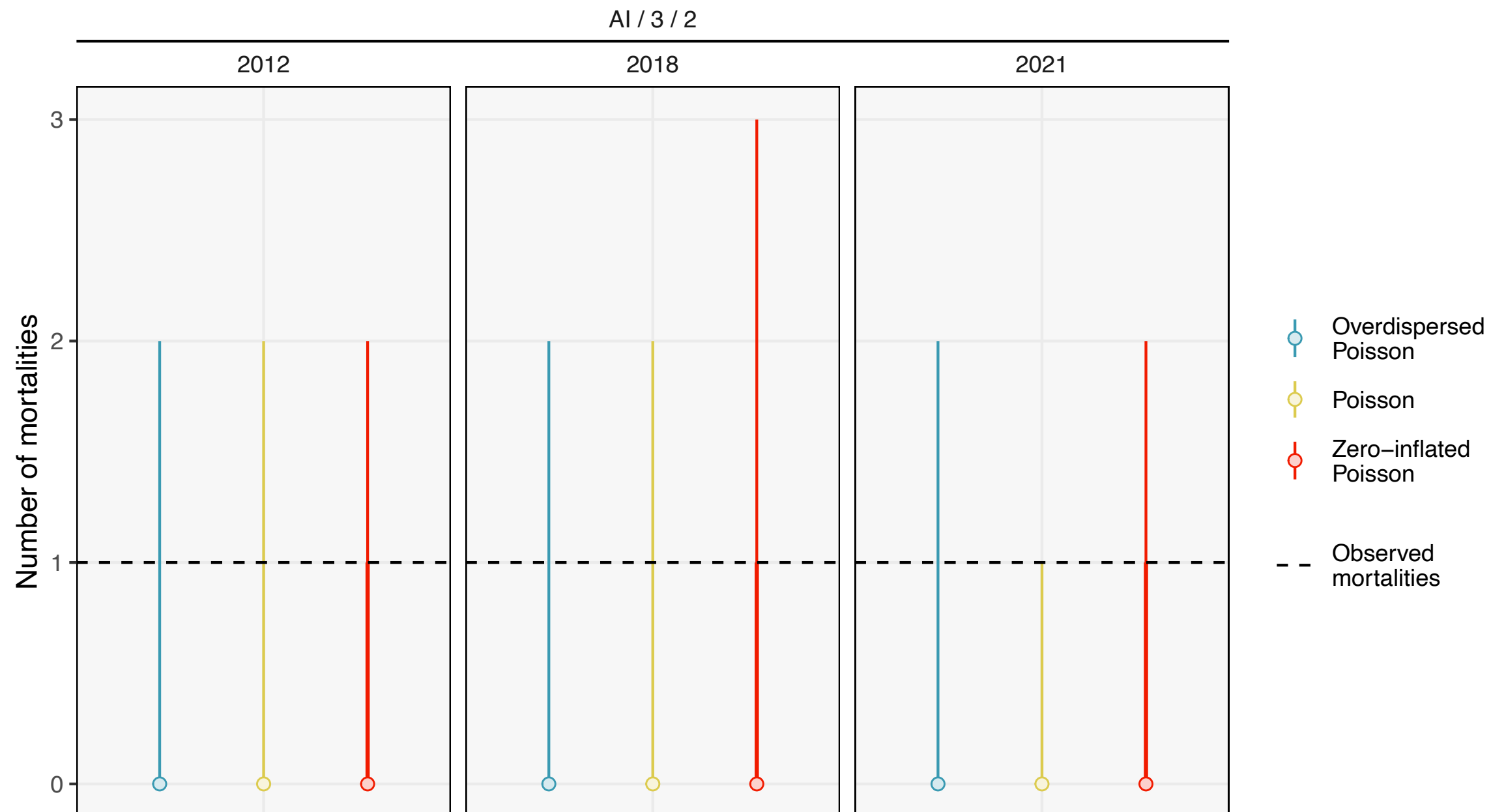
Model assessment

- Leave-one-out cross-validation: assess predictive ability using withheld data
- Compare with 2 other models: Poisson and zero-inflated Poisson
- ~4K models fit, 1 per strata per species, stock, and fishery
 - Out-of-sample coverage probability
 - MSE of out-of-sample predictions

Out-of-sample predictions

Northern elephant seal / California Breeding: BSAI Atka mackerel trawl (32)

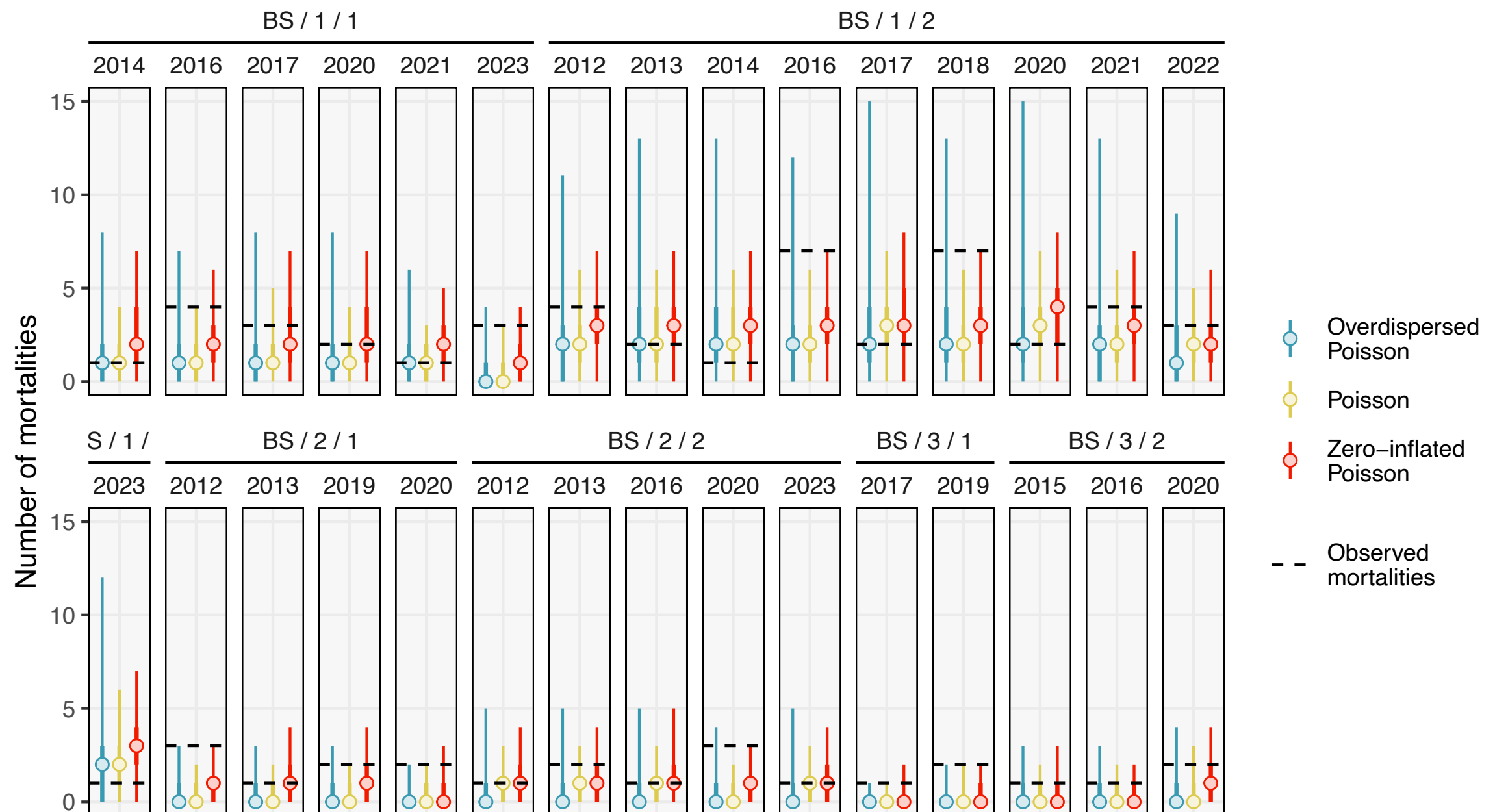
Total number of strata: 12



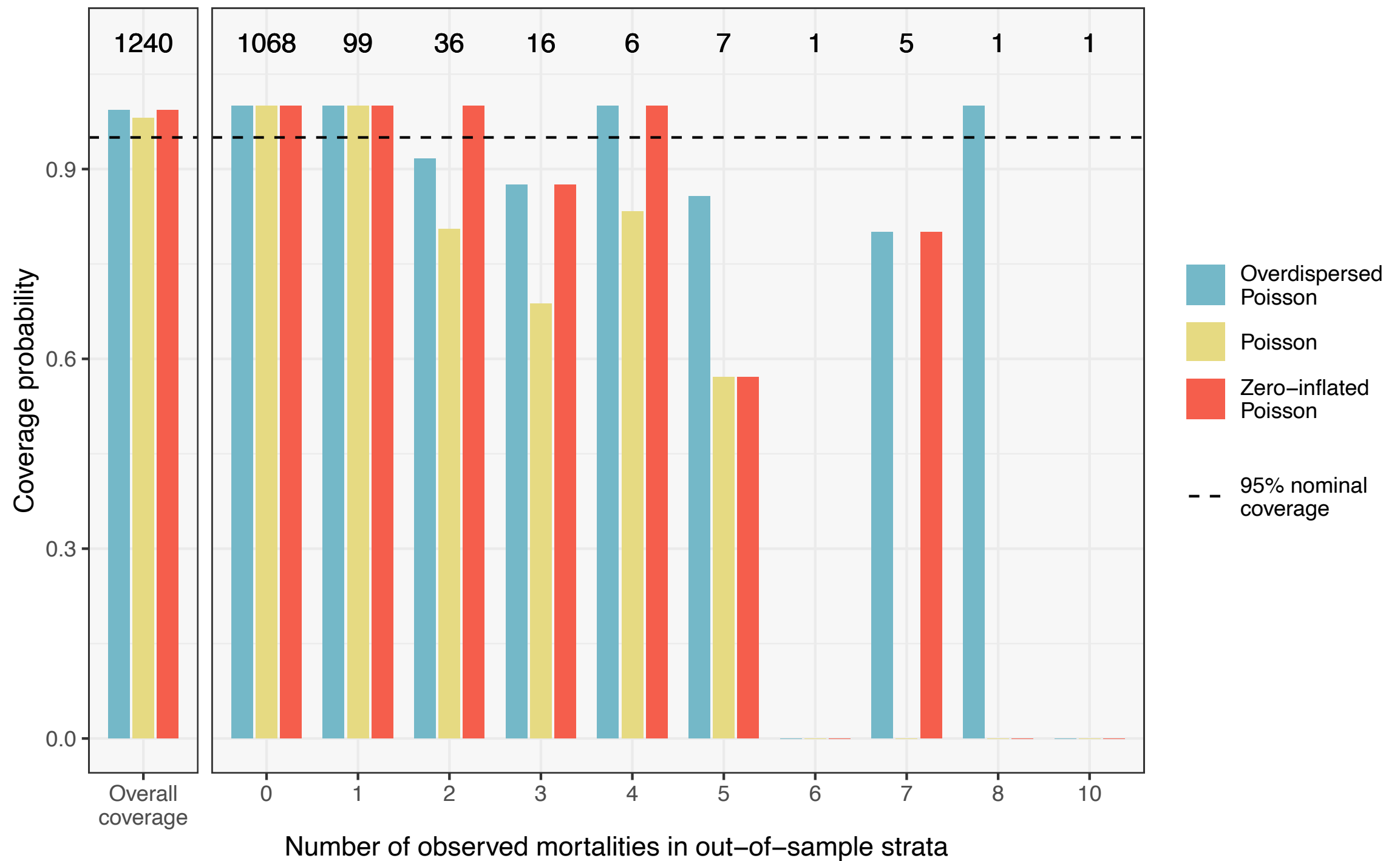
Out-of-sample predictions

Steller sea lion / Western: BSAI pollock trawl (56)

Total number of strata: 72



Out-of-sample coverage probability



Out-of-sample prediction error

